

SECTION B

7 (**short**) A metal is subjected to a Vickers hardness test as sketched in Fig. 7. The test resulted in an indent size $d = 0.48$ mm at a constant load $F = 5$ kg.

(a) Define the Vickers hardness and estimate the metal's yield stress σ_y . [4]

(b) The metal deforms in a standard tensile test according to the following relationship between true stress σ_t and true strain ϵ_t :

$$\sigma_t = K \epsilon_t^{0.5} \text{ MPa}$$

Assume the average true strain in a Vickers hardness test is 8%. Use the results from part (a) to calculate the constant K . Find the change in length of a tensile test specimen of initial length 20 cm, loaded with $\sigma_t = 200$ MPa. [6]

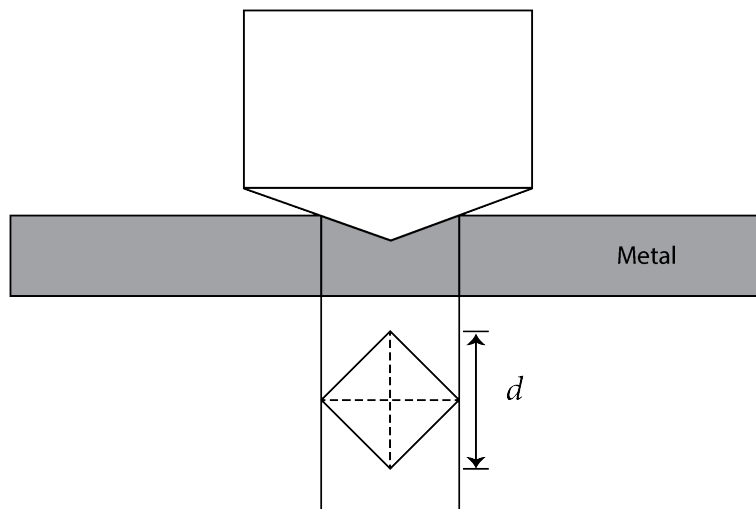


Fig. 7